

TEAM POCKETPILOT · VE411



PocketPilot

A privacy-aware agent orchestrator. It handles real tasks across your phone apps seamlessly, puts idle home compute to work, and keeps your raw data inside a trusted boundary.

Phone · local vault

Home · private tools

Cloud · assistant

TEAM NAME & MEMBERS

Team PocketPilot

ZZ

Zhaojiacheng Zhou

agent orchestration · inference

TJ

Tianqing Jiao

home-edge backend · router

YW

Yixiang Wang

android · UI automation

HZ

Haoxuan Zeng

UX research · market analysis

How this mockup was made

- ◇ **Design in Figma**; the clickable prototype is then a **hand-coded HTML · CSS · JavaScript** app that mirrors the Figma storyboard.
- ◇ **Interactive**: every button, tab and the “Anon” pill transitions to another screen — as required for a digital prototype.
- ◇ **Exported** to this PDF with Google Chrome (print-to-PDF). Font: Plus Jakarta Sans.

LIVE INTERACTIVE PROTOTYPE

<https://chhc-l.github.io/PocketPilot-Prototype/>

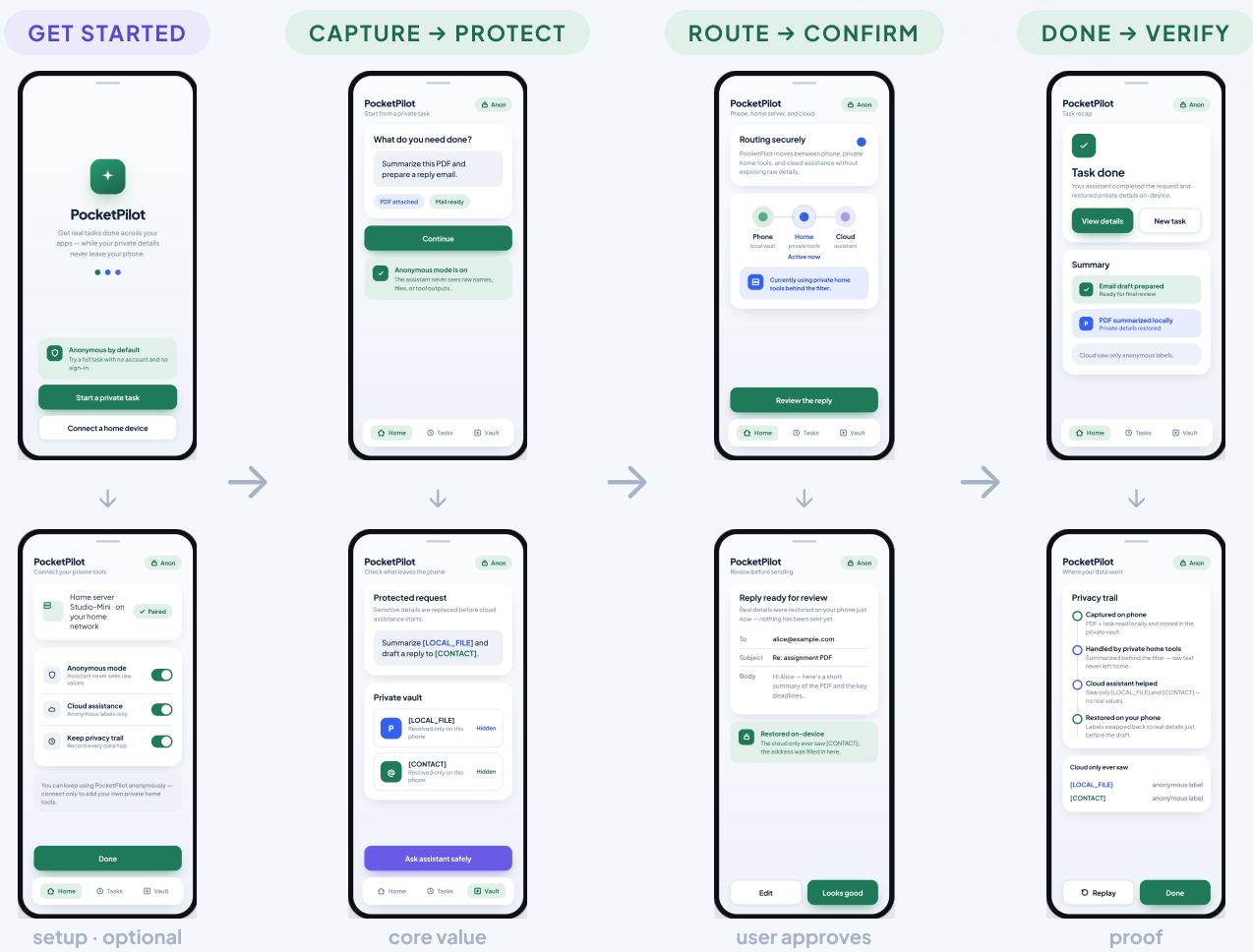
Public link — open in any browser. Staff contact: ve441staff.2025@gmail.com.

USABILITY PILOT-RUN VIDEO

<https://pan.sjtu.edu.cn/web/share/7cf0f7c153d80c4ffc0956986c8aabd7>

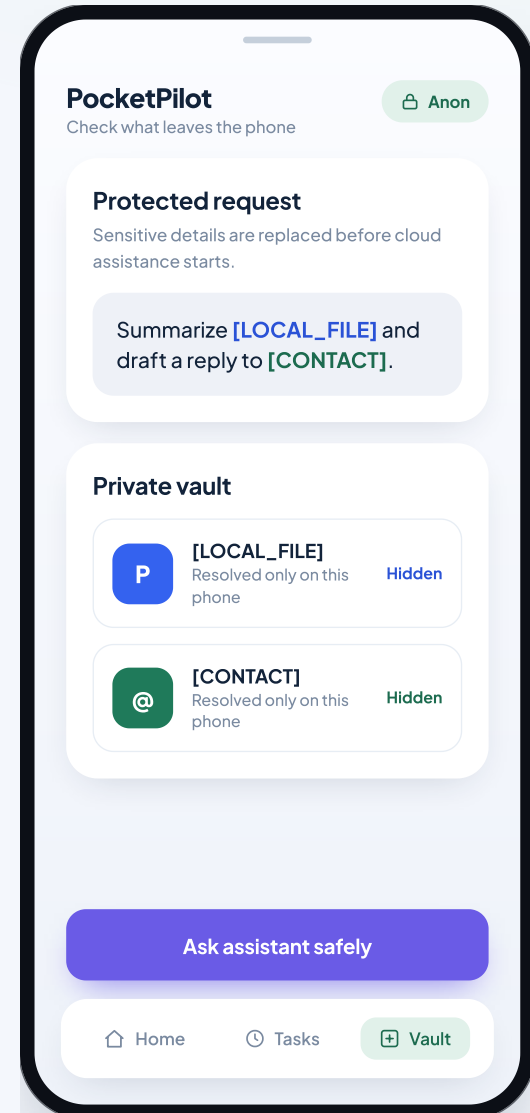
record.mp4 · pilot run only · ≤ 20 min.

One task, end to end — with the privacy trail visible

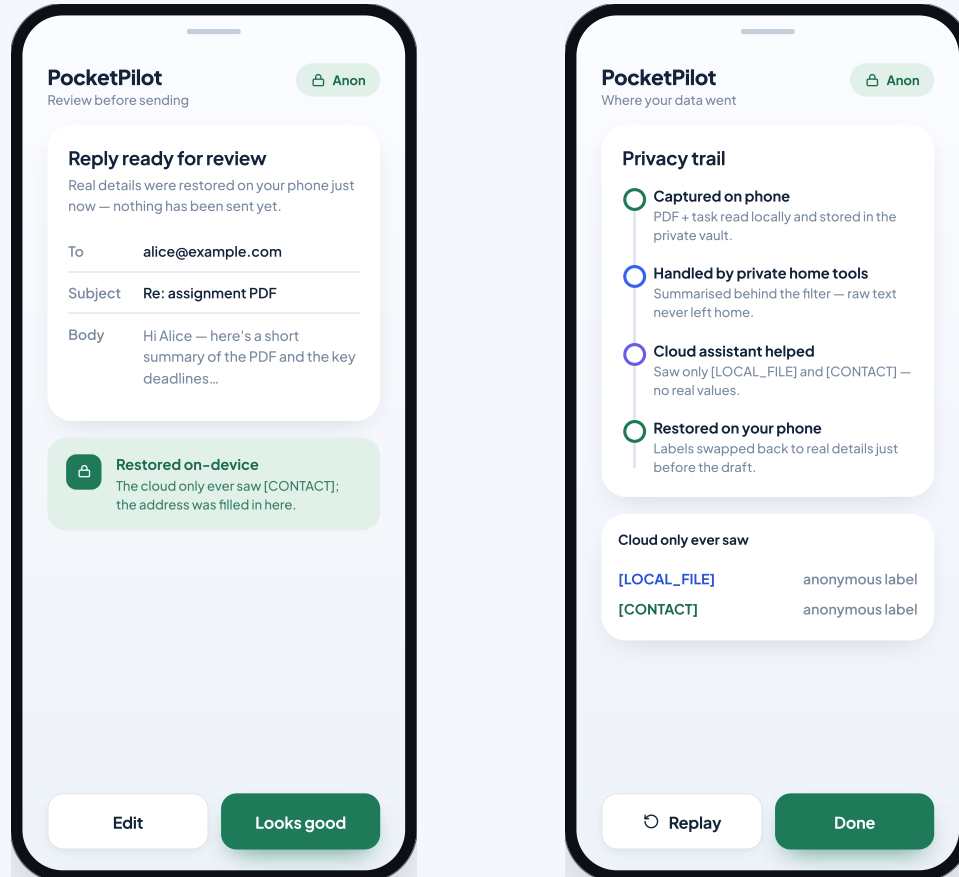


Privacy review: what leaves the phone is the user's choice

- ◆ **Protected request** — the task is rewritten as [LOCAL_FILE] and [CONTACT] before any cloud assistance starts.
- ◆ **Private vault** holds the real values, “resolved only on this phone”. They are never uploaded.
- ◆ **Branch:** *Ask assistant safely* proceeds; the user can also go back to reword or detach the file first.
- ◆ This screen carries the core value proposition: **capability without exposure**.



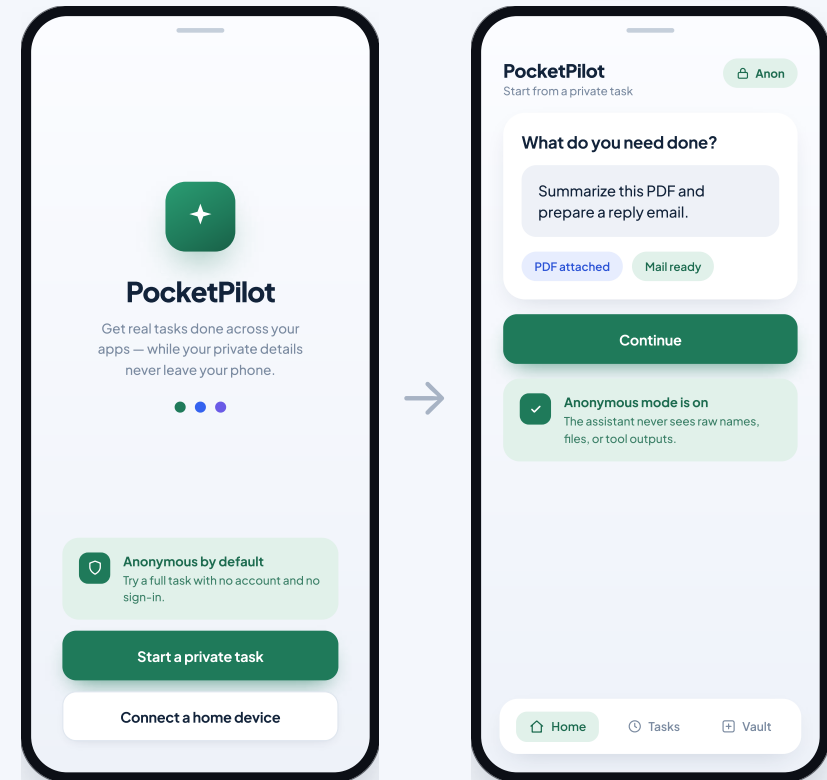
Confirm before sending → then prove where data went



- ◆ **Confirmation gate:** nothing is sent until the user approves. Real details are restored on-device only for this preview.
- ◆ **Edit branch** returns to the assistant with nothing sent.
- ◆ **Privacy trail** lists every hop and shows “cloud only ever saw anonymous labels” — the differentiator users can verify, not just trust.

Experience the value before any setup

- ✦ **Anonymous by default.** Setup is only needed to connect a home device, so we gate it as late as possible.
- ✦ **Start a private task** drops the user straight into a workspace with a sample file attached — no account, no sign-in wall.
- ✦ A full task (capture → protect → route → confirm → done → verify) runs entirely in anonymous mode.
- ✦ **Connect a home device** is the alternate branch for users ready to add their own private tools — the only path that reaches setup.

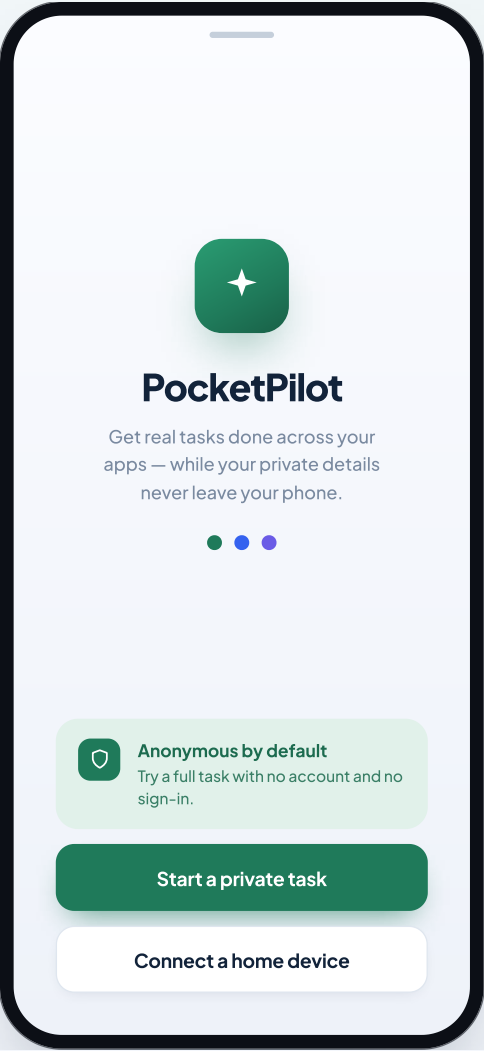


01

Welcome / trial entry

Anonymous by default — start a real task immediately, with no account and no device pairing required.

Get started 390 x 844



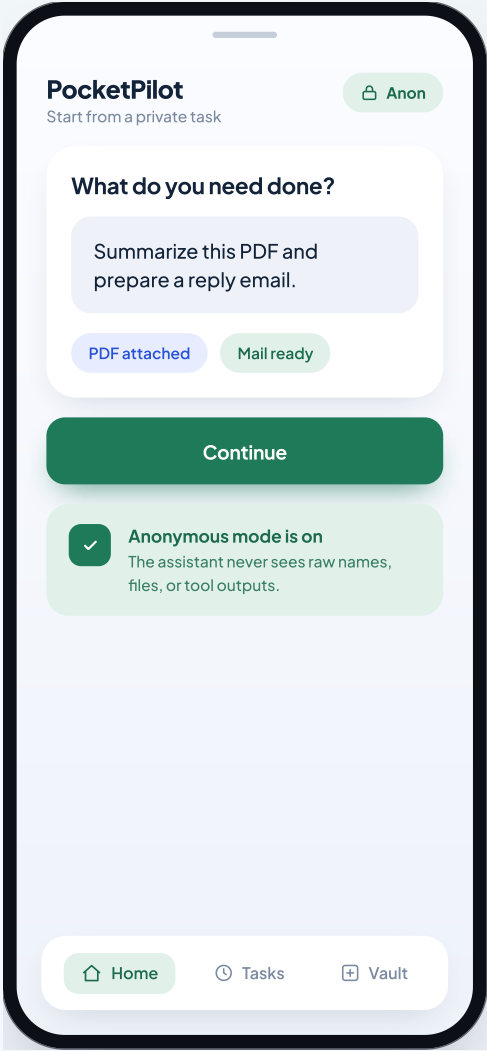
02

Task input

Describe any task in plain language. In demo mode a file is already attached and mail is ready, so users go straight to core value.

Capture

390 x 844



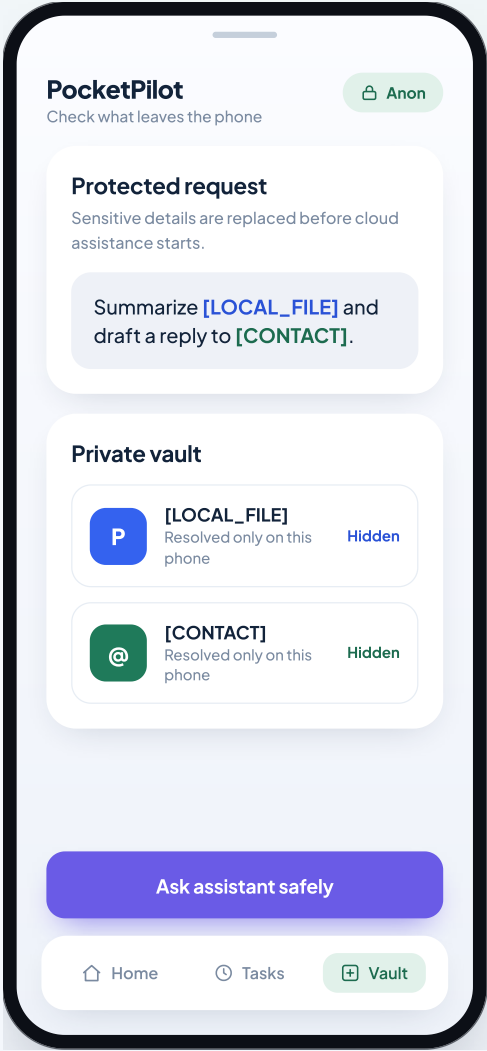
03

Privacy review

The core screen: the request is rewritten with [LOCAL_FILE] / [CONTACT] before anything reaches the cloud, and the private vault keeps the real values on-device.

Protect

390 × 844

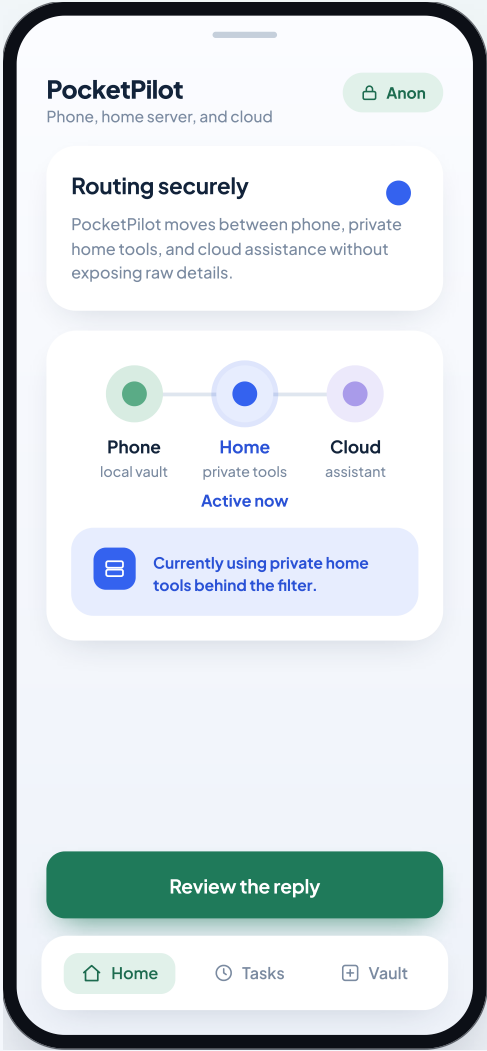


04

Routing securely

A live view of the task moving Phone → Home → Cloud, using private home tools behind the filter without exposing raw details.

Route 390 × 844



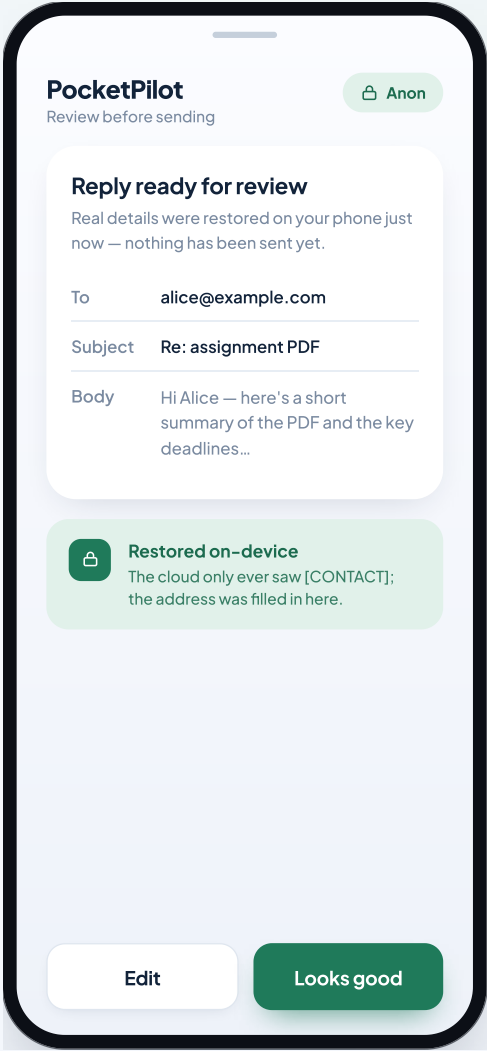
05

Review before sending

The confirmation gate: real details are restored on-device and the drafted reply is shown for approval before anything is sent.

Confirm

390 × 844



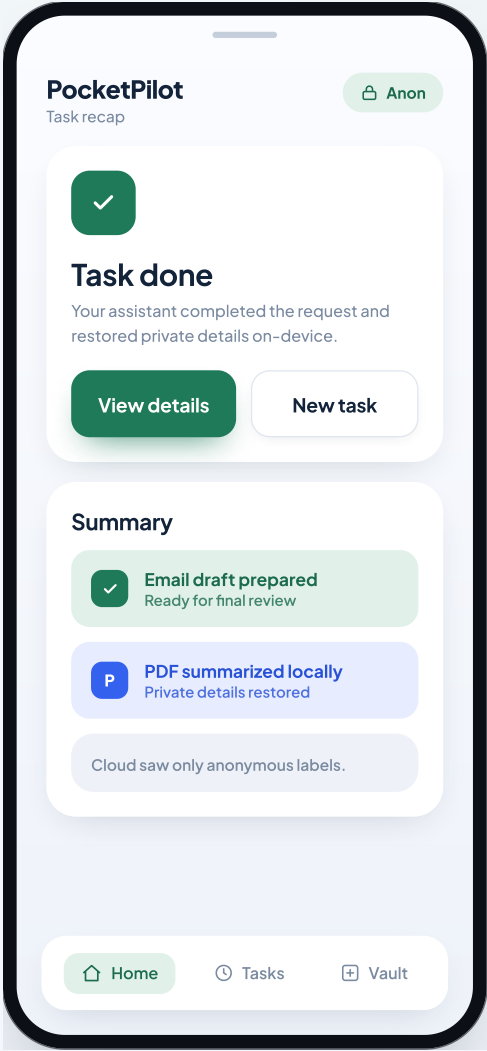
06

Task recap

Task done — the assistant finished and restored private details on-device; the summary shows the cloud saw only anonymous labels.

Done

390 × 844



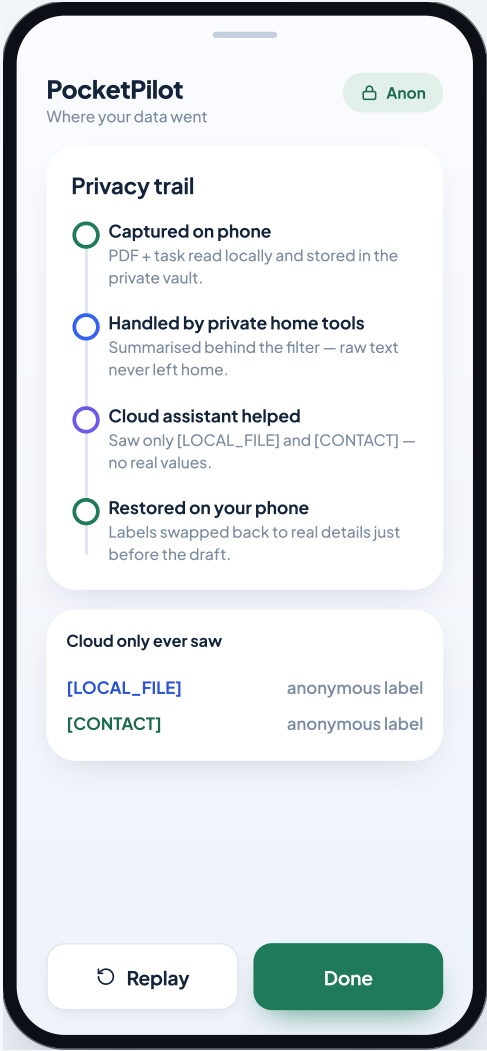
07

Privacy trail

A per-hop trail proving exactly what each domain saw — verifiable evidence that the cloud only ever received anonymous labels.

Verify

390 × 844

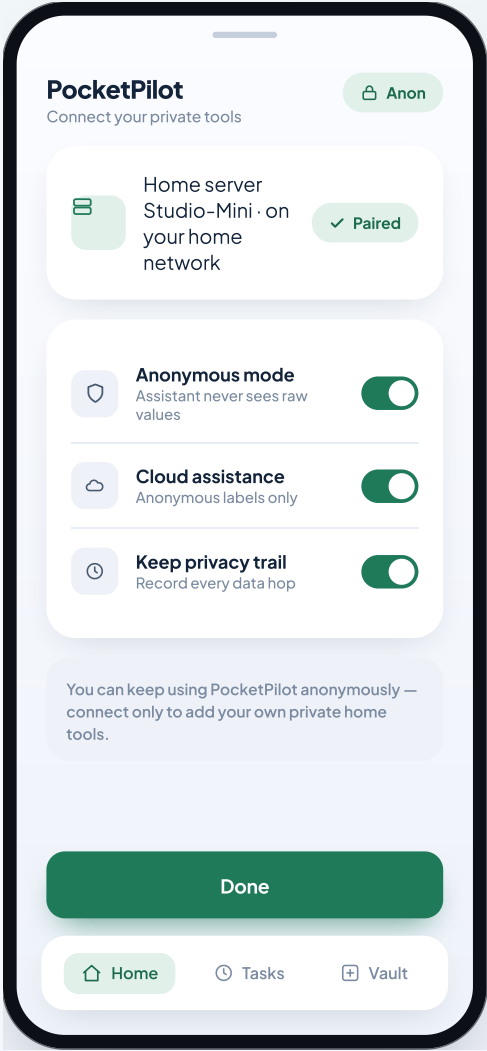


08

Connect private tools

The one place setup is needed: pair a home server to add your own private tools. Everything else works anonymously first.

Setup 390 x 844



What the facilitator says first

“Thanks for helping us out today. Before we start, a few things I want to be clear about:

- We're testing **the app, not you** — there are no wrong moves, and anything you find confusing is useful feedback for us.
- You can **stop at any time**, for any reason, and you don't have to explain why.
- Please **think out loud** as you go — tell me what you're looking at, what you expect to happen, and where you get stuck. **Ask questions** freely, though I may not answer some so I can see what you'd do on your own.
- We've already **created a test account and signed you in**, with a sample file loaded, so you can focus on the tasks.

I'll give you a few short tasks. Ready when you are.”

What the test tries to answer

Derived from our value proposition. We never ask testers these directly — we infer answers from how they perform the tasks.

- **RQ1 — Task expression:** Can a first-time user state a real task in plain language and trust PocketPilot to carry it out?
- **RQ2 — Privacy transparency:** Do users understand what stays on their phone versus what the cloud sees, and does that build trust?
- **RQ3 — Verifiability:** Can users confirm for themselves that sensitive details (a file, a contact) were only ever anonymous labels to the cloud?
- **RQ4 — Control:** Do users feel in control at the confirmation step, and can they stop or edit before anything is sent?
- **RQ5 — Onboarding friction:** Can users reach and complete core value in anonymous mode before any account or device pairing?

Tasks (given without telling users how)

- **T1 (RQ1):** “You have a PDF and want the app to summarise it and prepare a reply email to your groupmate. Get PocketPilot to do it.”
- **T2 (RQ2):** “Before the assistant does anything, find out what information about you leaves your phone, and what the cloud can see.”
- **T3 (RQ3):** “After the task finishes, check whether your groupmate's real details were ever sent to the cloud.”
- **T4 (RQ4):** “The reply isn't quite right — change it before it's sent.”
- **T5 (RQ5):** “You just installed PocketPilot and haven't set anything up. Try what it can do before creating an account or connecting a device.”

Tasks assume the tester is already signed in to a prepared test account.

Metric & expected value per task

TASK	SUCCESS CRITERION	METRIC	EXPECTED VALUE
T1 Run task	Reaches the “Task done” recap with a prepared reply	time · taps	≤ 45 s · ≤ 5 taps
T2 Find privacy view	States ≥1 item kept local & ≥1 cloud-visible	time to locate	≤ 20 s
T3 Verify no leak	Finds the privacy trail, answers “no” correctly	time · correctness	≤ 30 s · 100%
T4 Edit before send	Uses the confirm screen to edit before sending	taps · success	≤ 2 taps · 100%
T5 Demo before setup	Completes a task with 0 forced sign-ins	forced sign-ins	0

Expected values are the best achievable on the current design.

Screening & de-brief

Pre-test — classification only (no tutorial)

- Age range and smartphone OS you use most.
- How often do you use voice / AI assistants (Siri, ChatGPT, ...)?
- Have you ever done a multi-app chore by hand (e.g. read a file, then email someone about it)?
- How concerned are you about apps sending your data to the cloud? (1-5)
- Do you have a technical / CS background? (y/n)

Post-test — de-brief

- I understood what left my phone and what stayed on it. (1-5)
- I trusted PocketPilot to carry out the task. (1-5)
- The confirmation step made me feel in control. (1-5)
- What, if anything, was confusing or unexpected? (open)
- Would you use this for real tasks? Why / why not? (open)